CROP WEATHER CALENDARS

FOR

MAHARASHTRA

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PREFACE

Information on normal weather conditions and water requirement of crops is of vital importance for effective crop planning and management. From this point of view, the India Meteorological Department had compiled Crop Weather Calendars for major cereals, pulses and oilseed crops in the country on a districtwise basis. These calendars provide numerical as well as pictorial information on factors favourable or unfavourable to crops at various stages of their growth.

As and when new strains are introduced by plant breeders and bio-technologists, agricultural practices change accordingly, the crop weather calendars tend to lose some of their relevance. A periodical updating of these calendars thus becomes necessary. This publication which is the eleventh in the new series of crop weather calendars is for the state of Maharashtra for Jowar, Wheat, Cotton, Groundnut, Gram, Paddy, Safflower, Bajra, Tur, Sunflower, Soyabean and Sugarcane crops.

It is hoped that the crop weather calendars would prove useful in a wide range of applications such as plant breeding, crop adaptation, drought proofing and irrigation management, ultimately resulting in maximizing the yield.

The preparation of the calendars would not have been possible without the close co-operation of the Department of Agriculture, Govt. of Maharashtra which is gratefully acknowledged. These calendars were prepared by the Division of Agricultural Meteorology, Meteorological Office, Pune. I have great pleasure in acknowledging the work done by Shri S. K. Banerjee, DDGM (Agrimet), Dr H. P. Das, Director; Shri S. Sengupta, Meteorologist Grade I; Shri A. W. Mehandale, Assistant Meteorologist Grade I; Smt A. A. Kale, Smt S. H. Bhandari and Smt V. J. Patwardhan in preparing these calendars.

(R. R. KELKAR)
Director General Of Meteorology

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Introduction

1. General

Maharashtra is located in the north centre of Peninsular India and the dominant physical characteristics of the state is plateau character. The state is divided physically into two unequal parts; 1) coastal strip-Konkan and 2) Deccan plateau. Its western parts is bounded by the Sahyadri range and runs nearly parallel to the sea coast from north to south at a distance of 40-60 kilometers inland from the Arabian Sea. Satpura ranges are in the north where thick forest are found in Chandrapur and Bhandara districts in northeast while Ajanta and Satmala ranges run through central part of the state. It is bounded by Gujarat and Madhya Pradesh on the northern side, Karnataka and Andhra Pradesh on its southern side while Chattisgarh covers the eastern boundary of the state. The Western Ghats play a significant role in the monsoon circulation over the state which gives abundant rainfall along the coast and on the Ghats but eastern side has pockets of rain shadow areas. The state comprising of 35 districts with total geographical area of 3,07,713 square kilometers is the third largest state with population of 91.11 million as per 1991 census and are divided into four meteorological subdivisions, namely:

1) Coastal region called Konkan which consists of the districts of Mumbai, Raigad, Ratnagiri, Sindhudurg, Thane ;

2) Madhya Maharashtra comprising of the districts of Ahmednagar, Dhule, Jalgaon, Kolhapur, Nandurbar, Nasik, Pune, Sangli, Satara, Sholapur ;

3) Marathwada consists of Aurangabad, Beed, Jalna, Latur, Nanded, Osmanabad, Parbhani ;

4) Vidarbha consists of Akola, Amravati, Bhandara, Buldana, Chandrapur, Gadchiroli, Gondia, Nagpur, Wardha, Washim, Yeotmal.

The state is also divided into three natural regions;

a) Coastal areas known as Konkan and is characterized by rugged and hilly topography. Here heavy rainfall and steep slope cause severe soil erosion. A number of rivers and streams run through this area and join in the Arabian Sea.

b) Sahyadri hill ranges known as Western Ghats which is responsible for heavy rainfall in the coastal plains.

c) Deccan plateau is a land bounded by Satpura and Satmala ranges in the north, Sahyadri Ghats in the west. The valley area is flat and number of big rivers like Godavari and Krishna drain this area.

2. Climate

The climate is tropical and its major portion is semi-arid with 3 distinct seasons, viz. Rainy (June to September), Winter (November to February) and Summer (March to May). The major portion of annual rainfall (50-100cm) is received in 60-70 days and is confined to southwest monsoon.

3. Soils

There is a narrow strip of coastal alluvial soil in coastal districts, followed by laterite soils in Ratnagiri district, while in the remaining portion of Kolhapur and Thane districts, red loamy soils prevail. Black soils, mostly medium, occur in the rest of the state, with a few patches of deep black soils in south Nasik and Aurangabad districts and in Ahmednagar district extending to Parbhani and Nanded, Dhule and Jalgaon have significant patch of black cotton soil.

4. Temperature

Normal maximum temperature varies from 24-32^{0}C in January, 29-38^{0}C in March going up to 43^{0}C in May, the hottest month of the year. Maximum temperature is lowest in Mahabaleshwar and ranges between 19^{0}C and 29^{0}C. Minimum temperature is lowest in the month of December and January being 10-15^{0}C. The minimum temperature in June is between 22-28^{0}C. The annual mean temperature ranges from 20-28^{0}C.

5. Land use

The state has vast natural vegetation. About 60 % area is under crops and 21 % under forests. The irrigated area is about 16 % (3.6 mha) of the gross cropped area. About 61 % of the total workers in the state depend on agriculture and allied activities.

6. Rainfall

The annual rainfall varies from less than 50 cm in west Vidarbha to more than 200 cms in the coastal districts and adjoining areas. The districts lying between Vidarbha and Konkan receive rainfall between 50 to 200 cms. Rainfall is between 100-150 cm in the eastern Vidarbha region, 75 to 100 cm east of Aurangabad upto Nagpur.

7. Potential evapotranspiration (PE)

Potential evapotranspiration is the water transpired from a uniform, short, green, actively growing vegetation when water is unlimited. The concept assumes that there is ample supply of water at the site of evaporation and the rate is governed by evaporating capacity of the atmosphere. The highest values of potential evapotranspiration are over Madhya Maharashtra and Marathwada and are 190 cm annually, Jalgaon having the

highest value of 191 cm. In Vidarbha, annual potential evapotranspiration values are between 160 to 180 cm. In the southwest monsoon months of June and September, monthly potential evapotranspiration values ranges from 10 to 13 cm.

8. Cropping pat rn

In the conte of increasing production, it is necessary to examine the cropping patterns from a scientific angle and find out possible alternative patterns h ving higher potential. The cropping pattern primarily depends on the soil and climatic factors, but the evolution of cropping pattern in course of time is the combined effect of soil, climate, food habits a requirements and economic factors.

The important crops grown in the state are Jowar, Wheat Rice, Bajra, Groundnut, Tur, Gram, and Cotton. The area under production of these major crops of a typical year (1999 to 2000) in the state is given below:

Table

	Crops	Area (Thousand hectares)	Production (Thousand tonnes)
1.	Jowar	5,191	4,693
2.	Rice	1,520	2,559
3.	Wheat	1,049	1,436
4	Bajra	1,736	1,734
5.	Tur	1,041	868
6.	Gram	933	600
7.	Groundnut	540	572
8.	Cotton (lint)	3,254	527

9. Crop weather calendar

In preparation of crop weather calendars, information collected from state agricultural departments have been analysed in relation to agro-climatic factors.

There are critical periods in the life history of a crop from the date of sowing to the date of harvesting. Sowing time is also one of no small anxiety to the farmer. He knows that a heavy shower just after sowing will wash away the seed and seedlings and depress germination considerably. Again, if after sowing is completed, a prolonged drought follows, the tender seedlings would wither away necessitating resowing. It is well known that the weather at sowing time determines even the nature and variety of the crop that the farmer decides to grow in his field. If conditions are favourable during the first phase (germination and the early vegetative growth) the crop gets a good start. Thereafter, intervals of clear weather are required for interculture and weeding. During the period of rapid vegetative growth, the water requirement and need for sunshine for photosynthesis are maximum. Then follows the period of flowering and reproduction with little vegetative growth.

During the period of flowering and grain-formation, clear weather and minimum cloudiness are favourable. Cloudy wet weather during this period can ruin an otherwise promising crop. When the grains are forming and/or are in the milk stage, one day's hot wind can shrivel up the grains. Even after the entire season has been passed safely, the crop is not free from the danger of adverse and unexpected weather. A heavy shower of rain and hail can do incalculable damage to a standing crop ready for harvest. If warned in time, the farmer would hurry up his harvesting operations. Later, when the harvested crop is lying on the threshing floor, it is again on the mercy of any sudden shower. Here, also timely warning may help the farmer a lot.

In order to provide the farmer with an efficient weather service, it is essential that the weather forecaster is familiar with the farmer's special needs. The farmer should, on his part become familiar with the weather bulletins and learn how to interpret them. One of the function of Division of Agricultural Meteorology, Pune is to tell the farmer how the "Weather forecaster" can come to his aid and to tell the "Weather expert" on the other hand what the exact weather warning requirements of the farmers are. With this object in view, the detailed information regarding the major crops in different districts of the country, their date of sowing, the dates of commencement and duration of the important cultural operations and the important epochs in the life history of the crops during each season and their probable weather requirements have been collected from the agricultural departments in the various states.

The information collected have been condensed and represented in a pictorial form in what is called the "Crop Weather Calendars". The crop weather calendar consists of three parts. At the bottom of the calendar is shown diagramatically the life history of the crop from the date of sowing to the date of harvest and threshing. The important "phases" like sowing, germination, transplantation (in the case of rice), tillering, elongation, flowering, grain formation and harvest are also indicated. These "phases" cover certain time intervals indicated by horizontal arrows, which depend upon variations in (a) crop variety, (b) sowing dates from place to place and from year to year, and (c) in the nature of the crop itself. For ready reference, the months and standard weeks are marked at the bottom of the diagram.

The middle of the calendar shows the normal weather conditions of rainfall, temperature and sunshine hours. Normal phasewise water requirement are also indicated here.

The uppermost portion of the calendar indicates the nature of the weather warnings and the periods during which they are to be issued by the forecasting office concerned.

The weather has also another indirect control over the crops. Crops are liable to attacks of various diseases and pests. The intensity of their incidence is dependent upon certain weather conditions. These have also been depicted in the diagram.

It will be obvious that these diagrams are of immense help to the weather forecaster to decide, what warnings are to be issued for a particular district during a given weather situation and during a particular phase of the crop growth. These calendars are of equal interest to the agricultural and general public as well as various departments of Government concerned with agriculture and food production.

The homogeneous neighbouring districts having similar weather and soil conditions and growing the same type of crop have been clubbed together in preparing the diagrams. Information available from this is demonstrated with the help of the following illustration.

Illustration

The crop weather calendar on page 5 for a group of four districts viz. Kolhapur, Sangli, Latur and Satara for non-irrigated Jowar (kharif) crop is given as an example. The sowing periods in this group of districts varies from 22 to 29th weeks (28 May – 22 July), germination from 23 to 30th weeks (4 June – 29 July), vegetative growth from 27 to 33rd weeks (2 July – 19 Aug.) and flowering from 32 to 37th weeks (6 August – 16 September). Grain formation in these areas occurs from 37 to 42nd weeks (10 September - 21 Oct.) while harvesting is accomplished from 39th week onwards (from 24 September).

One of the critical phases of the crop is grain formation. During 40th week the average rainfall recorded is 36.3 mm. The mean daily maximum and minimum temperatures are 30.7^{0}C and 20.6^{0}C respectively while the average number of sunshine hours is 6.1. Rains of more than 100 mm normally in this period adversely affect the yield. Also maximum temperature > 30^{0}C and minimum temperature < 12^{0}C are harmful to crops. Warnings may be issued, if this type of adverse weather is anticipated.

Under favourable conditions of humid and rainy weather, pests like midge fly, stem borer and shoot fly may attack. Disease infestation like Helminthosporium, Ergot and Smut can also affect Jowar if low temperature accompanied by continuous rainfall prevails during the grain formation stage.

CONTENTS

Crop	Page No	Districts	Crop	Page No	Districts
Jowar	1	Buldhana, Jalna, Aurangabad, Parbhani	Groundnut	19	Kolhapur, Parbhani, Pune, Sindhudurg
	2	Amravati, Akola, Jalgaon	(Rabi)		
	3	Pune, Dhule, Beed, Osmanabad	Gram	20	Ahmednagar, Solapur, Kolhapur
	4	Wardha, Yavatmal, Nagpur		21	Wardha, Yavatmal, Nagpur, Parbhani
	5	Kolhapur, Sangli, Latur, Satara		22	Pune, Dhule, Thane
Jowar	6	Ahmednagar, Nanded, Beed, Pune	Paddy	23	Ratnagiri, Sindhudurg, Thane, Raigad
(Rabi)	7	Kolhapur, Satara, Solapur, Sangli	(Kharif)	24	Bhandara, Chandrapur, Gadchiroli, Nagpur
	8	Jalna, Aurangabad, Jalgaon, Parbhani	Safflower	25	Ahmednagar, Pune, Solapur
Wheat	9	Nagpur, Yavatmal, Amravati		26	Buldhana, Parbhani, Aurangabad
	10	Wardha, Bhandara, Chandrapur	Bajra	27	Dhule, Nasik, Ahmednagar, Aurangabad, Jalna
	11	Pune, Nasik, Dhule, Satara	(Kharif)	28	Pune, Satara, Sangli, Solapur
	12	Ahmednagar, Akola, Aurangabad, Buldhana	Tur/Arhar (Kharif)	29	Ahmednagar, Osmanabad, Solapur, Parbhani
				30	Buldhana, Amravati, Aurangabad, Nagpur
Cotton	13	Amravati, Wardha, Nagpur	Sunflower	31	Ahmednagar, Solapur
(Kharif)	14	Akola, Yavatmal, Nanded	(Kharif)		
	15	Jalna, Aurangabad, Parbhani	Soyabean	32	Amravati, Kolhapur, Nagpur
	16	Buldhana, Dhule, Jalgaon	(Kharif)		
Groundnut	17	Nagpur, Amravati, Dhule	Sugarcane	33	Pune, Satara
(Kharif)	18	Kolhapur, Sangli, Ahmednagar, Satara			

CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : BULDHANA, JALNA, AURANGABAD, PARBHANI

CROP : JOWAR (KHARIF) **NON-IRRIGATED**

VARIETY : CSH-9
DURATION : 110 - 120 DAYS

SOIL : HEAVY AND MEDIUM BLACK

Weather warnings	*Rain*	> 50 MM/DAY	> 65 MM/DAY	> 76 MM/DAY
	Duration of wet spell	25 MM FOR 3 DAYS	25 MM FOR 3 DAYS	30 MM FOR 3 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	15 DAYS	20 DAYS	20 DAYS
	High winds	> 40 KM/HR	> 40 KM/HR	> 40 KM/HR
	Temperature	MAX. TEMP. > 30°C, MIN.TEMP < 15°C	MAX. TEMP. > 35°C, MIN.TEMP. < 15°C	MAX. TEMP. > 35°C, MIN.TEMP. < 15°C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	SHOOT FLY	STEM BORER,APHIDS,ARMY WORM,MIDGE FLY	MIDGE FLY, ARMY WORMS
	Weather	CLOUDY WEATHER, HUMID CLIMATE	CLOUDY WEATHER, HUMID CLIMATE	CLOUDY WEATHER, HUMID CLIMATE
	Diseases	LEAF SPOTS, ERGOT	CHARCOAL ROT,DOWNY MILDEW,SMUT,MOULDS	
	Weather	HIGH TEMPERATURE, HIGH HUMIDITY	HIGH TEMPERATURE,HIGH HUMIDITY	

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : AMARAVATI, AKOLA, JALGAON

CROP : JOWAR (KHARIF)
NON-IRRIGATED

VARIETY : CSH-5,CSH-6,CSH-9,SPH 388 & SPV
DURATION : 110-120 DAYS

SOIL : MEDIUM TO DEEP BLACK

Weather warnings	*Rain*	> 100 MM/DAY	> 100 MM/DAY	> 100 MM/DAY
	Duration of wet spell	50 MM FOR 6 DAYS	75 MM FOR 8 DAYS	25 MM FOR 3 DAYS
	Cloudy weather			CLOUDY WEATHER
	Drought	15 TO 20 DAYS	15 TO 20 DAYS	15 TO 20 DAYS
	High winds			40-50 KM/HR
	Temperature	MAX.TEMP. > 40°C, MIN.TEMP < 15°C	MAX.TEMP. > 40°C, MIN.TEMP. < 10°C	MAX.TEMP. > 40°C, MIN.TEMP. < 12°C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	SHOOT FLY,STEM BORER,APHIDS,JASSIDS, SEEDLING CUTTING INSECTS,STEM FLY	ARMY WORMS,APHIDS,JASSIDS,STEM BORER, SUCKING TYPE PESTS	FLY WORMS,GRAIN EATING PESTS
	Weather	HIGH HUMIDITY,MODERATE TEMP . ABSENCE OF RAIN,CLOUDINESS	HIGH HUMIDITY,MODERATE TEMP,CONTINUOUS RAIN,CLOUDINESS	HIGH HUMIDITY,MODERATE TEMP DRY CONDITION
	Diseases	SUGARY DISEASE,BLIGHT,DOWNY MILDEW	SUGARY DISEASE,BLIGHT,RUST,CHARCOAL ROT	GRAIN MOULDS,GRAIN SMUT
	Weather	HIGH HUMIDITY,LOW SUNSHINE	HIGH TEMPERATURE	RAIN AT MATURITY

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : PUNE, DHULE, BEED, OSMANABAD

CROP : JOWAR (KHARIF) **NON-IRRIGATED**

VARIETY : CSH-1,CSH-5,CSH-6,CSH-9
DURATION : 110 - 120 DAYS

SOIL : LIGHT TO MEDIUM AND DEEP BLACK

Weather warnings	*Rain*	> 200 MM/DAY	> 200 MM/DAY	> 100 MM/DAY
	Duration of wet spell	25 MM FOR 5 DAYS	25 MM FOR 15 DAYS	50 MM FOR 8 DAYS
	Cloudy weather			CLOUDY WEATHER
	Drought	10 DAYS	20 DAYS	25 DAYS
	High winds			
	Temperature	MAX. TEMP. > 30^{0}C, MIN.TEMP. < 20^{0}C	MAX. TEMP. > 35^{0}C, MIN.TEMP. < 25^{0}C	MAX. TEMP. > 30^{0}C, MIN.TEMP. < 20^{0}C
	Hail Storm			HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	SHOOT FLY	MIDGE FLY, STEM BORER, ARMY WORM	MIDGE FLY, STEM BORER, ARMY WORM
	Weather	CLOUDY WEATHER, CONTINUOUS RAIN	CLOUDY WEATHER,CONTINUOUS RAIN	CLOUDY WEATHER, CONTINUOUS RAIN
	Diseases		DOWNY MILDEW, RUST	DOWNY MILDEW, ERGOT, SMUT
	Weather	CLOUDY WEATHER	CLOUDY WEATHER	CLOUDY WEATHER

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : WARDHA, YAVATMAL, NAGPUR

CROP : JOWAR (KHARIF) **NON-IRRIGATED**

VARIETY : CSH-5, CSH-9, AKSH 14 -150
DURATION : 110 - 115 DAYS

SOIL : HEAVY MEDIUM BLACK TO LIGHT, RED LATERITE

Weather warnings	*Rain*	> 75 MM/DAY	> 75 MM/DAY	> 25 MM/DAY
	Duration of wet spell	50 MM FOR 4 DAYS	50 MM FOR 7 DAYS	25 MM FOR 7 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	7 DAYS	14 DAYS	14 DAYS
	High winds			
	Temperature	MAX. TEMP. > 35^{0}C, MIN.TEMP < 15^{0}C	MAX. TEMP. > 35^{0}C, MIN.TEMP < 15^{0}C	MAX. TEMP. > 40^{0}C, MIN.TEMP. < 15^{0}C
	Hail Storm		HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	SHOOT FLY, APHIDS, JASSIDS	STEM BORER, ARMY WORM, APHIDS, JASSIDS	MIDGE FLY, HAIRY CATERPILLAR, HELIOTHIS
	Weather	EXCESS RAINS, HUMID AND CLOUDY WEATHER	EXCESS RAINS, HUMID AND CLOUDY WEATHER	EXCESS RAINS, HUMID AND CLOUDY WEATHER
	Diseases	LEAF SPOT	LEAF BLIGHT, RUST	ERGOT, SMUT, GRAIN MOULD
	Weather	EXCESS RAINS, HUMID AND CLOUDY WEATHER	EXCESS RAINS, HUMID AND CLOUDY WEATHER	EXCESS RAINS, HUMID AND CLOUDY WEATHER

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : KOLHAPUR, SANGLI, LATUR, SATARA

CROP : JOWAR (KHARIF) **NON-IRRIGATED**

VARIETY : CSH-1, CSH-5, CSH-9, MAHIKO-51
DURATION : 110 - 125 DAYS

SOIL : LIGHT TO MEDIUM AND HEAVY BLACK

Weather warnings	*Rain*	> 50 MM/DAY	> 100 MM/DAY	> 100 MM/DAY
	Duration of wet spell	50 MM FOR 8 DAYS	75 MM FOR 8 DAYS	50 MM FOR 10 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	6 DAYS	10 DAYS	15 DAYS
	High winds			
	Temperature	MAX. TEMP. > 31°C, MIN.TEMP. < 12°C	MAX. TEMP. > 31°C, MIN.TEMP. < 12°C	MAX. TEMP. > 30°C, MIN.TEMP. < 12°C
	Hail Storm			HAIL STORM
Weather conditions ourable for incidence of pests and diseases	*Pests*	STEM BORER, SHOOT FLY	STEM BORER, SHOOT FLY, ARMY WORM	MIDGE FLY, STEM BORER, SHOOT FLY ARMY WORM, EARHEAD BUG
	Weather	CLOUDY WEATHER	HIGH HUMIDITY	HIGH HUMIDITY, HIGH RAINFALL
	Diseases	HELMINTHOSPORIUM, ERGOT, SMUT	HELMINTHOSPORIUM, ERGOT, SMUT	HELMINTHOSPORIUM, ERGOT, SMUT
	Weather	LOW TEMPERATURE, CONTINUOUS RAINFALL, CLOUDINESS	LOW TEMPERATURE, CONTINUOUS RAINFALL, CLOUDINESS	LOW TEMPERATURE, CONTINUOUS RAINFALL, CLOUDINESS

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP :** JOWAR (RABI) **VARIETY** : M-35-1, SPV-86 **SOIL :** MEDIUM BLACK

DISTRICTS : AHMEDNAGAR, NANDED, BEED, PUNE **IRRIGATED/NON-IRRIGATED** **DURATION :** 110-120 DAYS

Weather warnings	*Rain*	> 100 MM/DAY	> 150 MM/DAY	> 150 MM/DAY
	Duration of wet spell	>10 MM FOR 20 DAYS	> 50 MM FOR 20 DAYS	> 70 MM FOR 20 DAYS
	Cloudy weather		CLOUDY WEATHER	
	Drought	8 DAYS	30 DAYS	30 DAYS
	High winds			
	Temperature	MAX.TEMP. > 30°C, MIN.TEMP. < 15°C	MAX.TEMP. > 30°C, MIN.TEMP. < 12°C	MAX.TEMP. > 30°C, MIN.TEMP. < 12°C
	Hail Storm		HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS, STEM BORER, SHOOT FLY, ARMY WORM, MIDGE FLY	APHIDS, STEM BORER, SHOOT FLY, ARMY WORM, MIDGE FLY	APHIDS, STEM BORER, SHOOT FLY, ARMY WORM, MIDGE FLY
	Weather	CLOUDY AND HUMID WEATHER	CLOUDY AND HUMID WEATHER	CLOUDY AND HUMID WEATHER
	Diseases	DOWNY MILDEW	SMUT, ERGOT	RUST
	Weather	CLOUDINESS	CLOUDINESS	CLOUDINESS

| **Normal phase wise water requirement(mm)** | | | | | | | | | | | **45.1** | | | | **60.4** | | | **99.4** | | | | **51.7** | | | **38.9** | | | | | | | | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : KOLHAPUR, SATARA, SOLAPUR, SANGLI

CROP : JOWAR (RABI)
NON-IRRIGATED

VARIETY : M-397, CSH-9, M-35-1, MALDANDI-35
DURATION : 110-130 DAYS

SOIL : LIGHT TO HEAVY BLACK

Weather warnings	*Rain*	> 65 MM/DAY	> 70 MM/DAY	> 80 MM/DAY
	Duration of wet spell			
	Cloudy weather			CLOUDY WEATHER
	Drought	25 DAYS	40 DAYS	30 DAYS
	High winds	> 40 KM/HR	> 15 KM/HR	> 10 KM/HR
	Temperature	MAX.TEMP > 40°C, MIN.TEMP. <16°C	MAX.TEMP. > 40°C, MIN.TEMP. <16°C	MAX.TEMP. > 40°C, MIN.TEMP. <16°C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	STEM BORER, SHOOT FLY	APHIDS, STEM BORER, JASSIDS	APHIDS, WEB WORM
	Weather	CLOUDY WEATHER, HIGH HUMIDITY	CLOUDY WEATHER, HIGH HUMIDITY	CLOUDY WEATHER, HIGH HUMIDITY
	Diseases	LEAF SPOT	RUST AND SMUT, CHARCOAL ROT	GREEN SMUT
	Weather	LOW TEMP., HIGH HUMIDITY	CLOUDY AND HIGH HUMIDITY	

| **Normal phase wise water requirement(mm)** | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : JALNA, AURANGABAD, JALGAON, PARBHANI

CROP : JOWAR (RABI)
NON-IRRIGATED

VARIETY : M-35-1,MALDANDI-35
DURATION : 120-140 DAYS

SOIL : MEDIUM TO HEAVY BLACK

Weather warnings	*Rain*	> 50 MM/DAY	> 65 MM/DAY	> 65 MM/DAY
	Duration of wet spell	40 MM FOR 3 DAYS	50 MM FOR 3 DAYS	50 MM FOR 3 DAYS
	Cloudy weather	CLOUDY WEATHER	CLOUDY WEATHER	CLOUDY WEATHER
	Drought	15 DAYS	20 DAYS	20 DAYS
	High winds			
	Temperature	MAX.TEMP. > 30^{0}C, MIN.TEMP. < 10^{0}C	MAX.TEMP. > 35^{0}C, MIN.TEMP. < 15^{0}C	MAX.TEMP. > 35^{0}C, MIN.TEMP. < 15^{0}C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	FLEA BEETLE, APHIDS, JASSIDS	APHIDS, JASSIDS	APHIDS, JASSIDS
	Weather	HUMID CLIMATE	CONTRAST TEMP. DURING DAY AND NIGHT	HIGH HUMIDITY
	Diseases	SMUT	VARIOUS LEAF SPOTS	VARIOUS LEAF SPOTS, RED STRIPE, ERGOT
	Weather	WARM, HUMID CLIMATE	WARM, HUMID CLIMATE	WARM, HUMID CLIMATE

| **Normal phase wise water requirement(mm)** | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP** : WHEAT **VARIETY** : HD-2189, 4502, 2273, 977, 2278, 1593, AHW-3181,SONALIKA, NI-5439,HDM-1552 **SOIL** : HEAVY TO MEDIUM BLACK & RED LATERITE

DISTRICTS : NAGPUR, YAVATMAL, AMARAVATI **IRRIGATED** **DURATION** : 105-125 DAYS

Weather warnings	*Rain*	> 100-150 MM/DAY	> 100-150 MM/DAY	> 100-150 MM/DAY
	Duration of wet spell	> 20 MM FOR 20 DAYS	> 20 MM FOR 20 DAYS	> 20 MM FOR 20 DAYS
	Cloudy weather	CLOUDY WEATHER	CLOUDY WEATHER	CLOUDY WEATHER
	Drought			
	High winds			
	Temperature	MAX. TEMP. > 30°C, MIN. TEMP. < 15°C	MAX. TEMP. > 30°C, MIN. TEMP. < 15°C	MAX. TEMP. > 32°C, MIN. TEMP. < 15°C
	Hail Storm			HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS, STEM BORER, TERMITES, JASSIDS	JASSIDS, APHIDS, STEM BORER	MITES, STEM BORER, APHIDS
	Weather	HIGH HUMIDITY, CLOUDINESS	HIGH HUMIDITY, CLOUDINESS	HIGH HUMIDITY, CLOUDINESS
	Diseases	WILT	BROWN SMUT, ALTERNARIA, HELMINTHOSPORIUM, LEAF SPOT	LEAF RUST, STEM RUST, SMUT
	Weather	MOIST WITH HIGH HUMIDITY	HIGH HUMIDITY, LOW TEMP.	CLOUDINESS

| **Normal phase wise water requirement(mm)** | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP** : WHEAT **VARIETY** : HD-1553,1593, 2278, 2189 **SOIL** : MEDIUM TO DEEP BLACK

DISTRICTS : WARDHA, BHANDARA, CHANDRAPUR **IRRIGATED** **DURATION** : 115-120 DAYS

Weather warnings	*Rain*	> 50 MM/DAY	> 75-100 MM/DAY	> 50 MM/DAY
	Duration of wet spell	> 50-75 MM FOR 3-4 DAYS	> 75-100 MM FOR 3-4 DAYS	> 50-75 MM FOR 3-4 DAYS
	Cloudy weather	CLOUDY WEATHER	CLOUDY WEATHER	
	Drought	7 DAYS	10 - 15 DAYS	8 - 10 DAYS
	High winds			
	Temperature	MAX. TEMP. > 30°C, MIN. TEMP. < 10°C	MAX. TEMP. > 25°C, MIN. TEMP < 13°C	MAX. TEMP. > 30°C, MIN. TEMP. < 13°C
	Hail Storm			HAIL STROM
Weather conditions favourable for incidence of pests and diseases	*Pests*	JASSIDS, TERMITES, STEM BORER, MITES		JASSIDS, TERMITES, MITES, STEM BORER
	Weather	CLOUDY WEATHER		CLOUDY WEATHER
	Diseases	BLIGHT, WILT, ALTERNARIA, HELMINTHOSPORIUM	RUST	LEAF RUST, STEM RUST, LOOSE SMUT
	Weather	COLD MOIST WEATHER WITH HIGH HUMIDITY	COLD MOIST WEATHER WITH HIGH HUMIDITY	COLD MOIST WEATHER WITH HIGH HUMIDITY

Normal phase wise water requirement(mm)		**23.1**	**65.2**	**121.1**	**73.3**	**28.8**

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP** : WHEAT **VARIETY** : HD - 2189, 2278, 4202, HDM-1593, 1553, NH-5439 **SOIL** : MEDIUM TO HEAVY BLACK

DISTRICTS : PUNE, NASIK, DHULE, SATARA **IRRIGATED** **DURATION** : 115-140 DAYS

Weather warnings	*Rain*	> 100-150 MM/DAY	> 100-150 MM/DAY	> 100-150 MM/DAY
	Duration of wet spell	> 20 MM FOR 20 DAYS	> 20 MM FOR 20 DAYS	> 20 MM FOR 20 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	20 DAYS	20 DAYS	20 DAYS
	High winds			
	Temperature	MAX. TEMP. > 30^{0}C, MIN. TEMP. < 15^{0}C	MAX. TEMP. > 30^{0}C, MIN. TEMP. < 10^{0}C	MAX. TEMP. > 30^{0}C, MIN. TEMP. < 20^{0}C
	Hail Storm			
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS, JASSIDS, MITES, STEM BORER	APHIDS, JASSIDS, MITES, STEM BORER	APHIDS, JASSIDS, MITES, STEM BORER
	Weather	CLOUDY WEATHER WITH HIGH HUMIDITY	CLOUDY WEATHER	CLOUDY WEATHER
	Diseases	RUST, ALTERNARIA	RUST, ALTERNARIA	RUST, ALTERNARIA
	Weather	HIGH HUMIDITY AND IRREGULAR RAINFALL	HIGH HUMIDITY AND IRREGULAR RAINFALL	HIGH HUMIDITY AND IRREGULAR RAINFALL

Normal phase wise water requirement(mm)		**33.7**	**88.4**	**131.0**	**106.1**	**31.9**	

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP :** WHEAT **VARIETY :** HD - 4502, 2189, 2278, 1593, 1553, HI- 977 **SOIL :** MEDIUM TO DEEP BLACK

DISTRICTS : AHMEDNAGAR, AKOLA, AURANGABAD, BULDHANA **IRRIGATED** **DURATION :** 115-120 DAYS

Weather warnings	*Rain*	> 30 MM/DAY	> 50 MM/DAY	> 50 MM/DAY
	Duration of wet spell	> 30 MM FOR 5 DAYS	> 30 MM FOR 5 DAYS	> 30 MM FOR 5 DAYS
	Cloudy weather		CLOUDY WEATHER	
	Drought	30 DAYS	30 DAYS	30 DAYS
	High winds	> 20 KM/HR		
	Temperature	MAX. TEMP. > 30°C, MIN. TEMP. < 15°C	MAX. TEMP. > 30°C, MIN. TEMP. < 15°C	MAX. TEMP. > 32°C, MIN. TEMP. < 15°C
	Hail Storm			
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS, JASSIDS,TERMITES	APHIDS, STEM BORER, THRIPS	MITES, STEM BORER, APHIDS
	Weather	CLOUDY WEATHER	CLOUDY WEATHER	CLOUDY WEATHER
	Diseases	BLACK ARM	RUST, ALTERNARIA, WILT, LEAF SPOT, HELMINTHOSPORIUM	RUST, ALTERNARIA, SMUT, LEAF RUST, STEM RUST
	Weather	CLOUDY WEATHER, HIGH HUMIDITY	CLOUDY WEATHER, HIGH HUMIDITY	CLOUDY WEATHER, HIGH HUMIDITY

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP :** COTTON (KHARIF) **VARIETY** : H4, H6, PKV-2, DHV-286, DESI-8401 **SOIL :** LIGHT, MEDIUM TO HEAVY BLACK AND HEAVY, MEDIUM TO LIGHT RED LATERITE SOIL

DISTRICTS : AMARAVATI, WARDHA NAGPUR **IRRIGATED/NON-IRRIGATED** **DURATION :** 140-210 DAYS

Weather warnings	*Rain*	>75 MM/DAY	>75 MM/DAY	>75 MM/DAY
	Duration of wet spell	50 MM FOR 8 DAYS	75 MM FOR 8-10 DAYS	50 MM FOR 8-10 DAYS
	Cloudy weather	CLOUDY WEATHER	CLOUDY WEATHER	CLOUDY WEATHER
	Drought	15 DAYS	15 DAYS	20 DAYS
	High winds	> 40 KM/HR	> 40 KM/HR	> 40 KM/HR
	Temperature	MAX.TEMP. > 30^{0}C, MIN.TEMP. < 15^{0}C	MAX.TEMP. > 35^{0}C, MIN.TEMP. < 15^{0}C	MAX.TEMP. > 40^{0}C, MIN.TEMP. < 15^{0}C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	JASSIDS, APHIDS, THRIPS, MITES	JASSIDS, APHIDS, THRIPS, MITES, WHITE FLY, LEAF ROLLER, SHOOT BORER	WHITE FLY, BOLL WORM, HELIOTHIS
	Weather	HIGH HUMIDITY AND CLOUDINESS	HIGH HUMIDITY AND CLOUDINESS	HIGH HUMIDITY AND CLOUDINESS
	Diseases	WILT, RED LEAF, BLACK ARM	POWDERY MILDEW, DOWNY MILDEW, BLACK ARM, LEAF BLIGHT, ANTHRAXNOSE	DOWNY MILDEW, BLAST
	Weather	HIGH HUMIDITY, CLOUDY WEATHER	HIGH HUMIDITY, CLOUDY WEATHER	HIGH HUMIDITY, CLOUDY WEATHER

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : AKOLA, NANDED, YAVATMAL

CROP : COTTON (KHARIF) **VARIETY** : HYBRID & DESHI (AHH-468,AKH-081) **SOIL** : MEDIUM TO HEAVY BLACK
IRRIGATED/NON-IRRIGATED **DURATION** : 150 - 200 DAYS

Weather warnings	*Rain*			
	Duration of wet spell	> 15-25 MM FOR 5 DAYS	> 29-50 MM FOR 5 DAYS	> 50-60 MM FOR 2 DAYS
	Cloudy weather		CLOUDY WEATHER	
	Drought	20 DAYS	18 DAYS	15 DAYS
	High winds	> 30 KM/HR	> 30 KM/HR	> 20 KM/HR
	Temperature	MAX. TEMP. > 40°C, MIN.TEMP < 16°C	MAX. TEMP. > 40°C, MIN.TEMP < 27°C	MAX. TEMP. > 40°C, MIN.TEMP. < 30°C
	Hail Storm		HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	JASSIDS, APHIDS, THRIPS, MITES, CATERPILLAR	JASSIDS, APHIDS, THRIPS, MITES, CATERPILLAR, HELIOTHIS, SPOTTED BOLL WORM	BOLLWORM, HELIOTHIS, PINK BOLL WORM, WHITE FLIES
	Weather	CLOUDY WEATHER, HIGH HUMIDITY	CLOUDY WEATHER, HIGH HUMIDITY	CLOUDY WEATHER, HIGH HUMIDITY
	Diseases	BLACK ARM, WILT, ROOT ROT, ANTHRAXNOSE	BLACK ARM, WILT, ROOT ROT, ANTHRAXNOSE	BLACK ARM, WILT, ROOT ROT, ANTHRAXNOSE
	Weather	HIGH HUMIDITY, MORE RAINY DAYS CLOUDY WEATHER	HIGH HUMIDITY, MORE RAINY DAYS CLOUDY WEATHER	HIGH HUMIDITY, MORE RAINY DAYS CLOUDY WEATHER

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : JALNA, AURANGABAD, PARBHANI

CROP : COTTON (KHARIF) **VARIETY :** DCM-32, NH-44, DESI COTTON **SOIL :** MEDIUM TO HEAVY BLACK
IRRIGATED/NON-IRRIGATED **DURATION :** 160-180 DAYS

Weather warnings	*Rain*	> 50 MM/DAY	> 65 MM/DAY	> 65 MM/DAY
	Duration of wet spell	40 MM FOR 3 DAYS	50 MM FOR 3 DAYS	50 MM FOR 3 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	15 DAYS	25 DAYS	20 DAYS
	High winds		>10 KM/HR	>10 KM/HR
	Temperature	MAX. TEMP. > 30°C, MIN.TEMP. < 15°C	MAX. TEMP. > 35°C, MIN.TEMP. < 15°C	MAX. TEMP. > 35°C, MIN.TEMP. < 15°C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS, JASSIDS, THRIPS, BOLL WORM	APHIDS, JASSIDS, THRIPS, BOLL WORM	APHIDS, JASSIDS, THRIPS, BOLL WORM
	Weather	WARM, HUMID CLIMATE	WARM, HUMID CLIMATE	WARM, HUMID CLIMATE
	Diseases	BLACK ARM, ALTERNARIA, LEAF SPOT	ANGULAR LEAF SPOT, POWDERY MILDEW	ANGULAR LEAF SPOT, POWDERY MILDEW
	Weather	WARM, HUMID CLIMATE	WARM, HUMID CLIMATE	WARM, HUMID CLIMATE

| **Normal phase wise water requirement(mm)** | | | |

CROP WEATHER CALENDAR

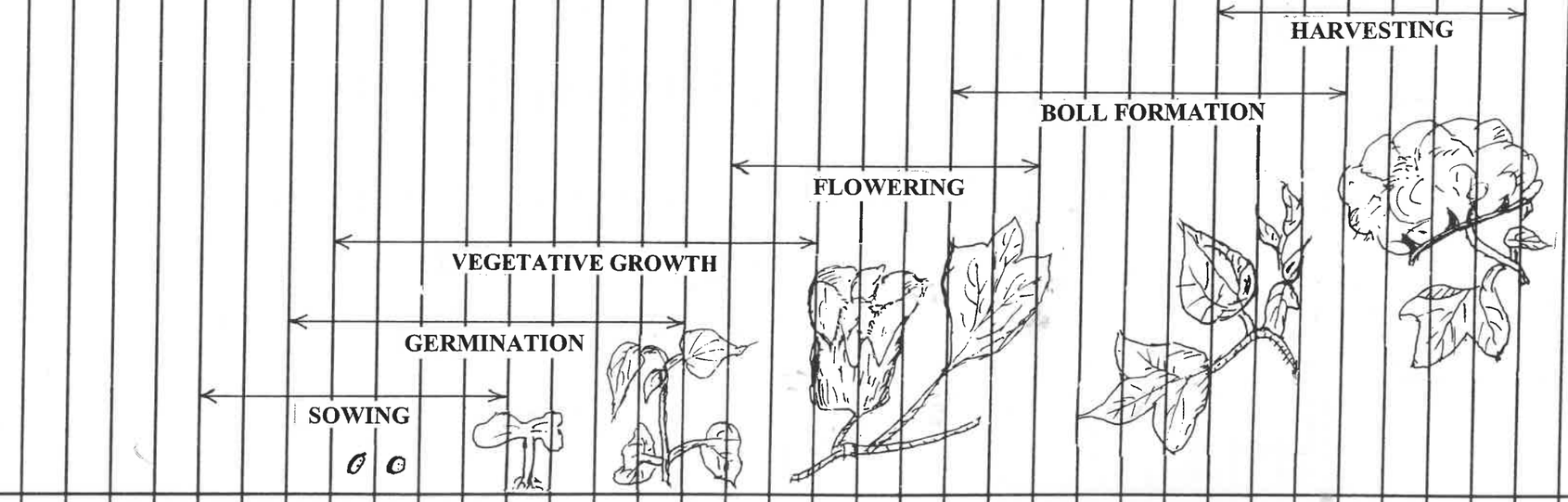
STATE : MAHARASHTRA **CROP :** COTTON (KHARIF) **VARIETY** : LRA-5166, L-147, H-4,6 **SOIL :** LIGHT, MEDIUM TO DEEP BLACK SOIL

DISTRICTS : BULDHANA, DHULE, JALGAON **IRRIGATED/NON-IRRIGATED** **DURATION :** 180-200 DAYS

Weather warnings	*Rain*	> 100 MM/DAY	> 100-125 MM/DAY	> 100-125 MM/DAY
	Duration of wet spell	40 MM FOR 20 DAYS	50 MM FOR 25 DAYS	40 MM FOR 25 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	21 DAYS	20 DAYS	20 DAYS
	High winds	> 40 KM/HR	> 40 KM/HR	> 40 KM/HR
	Temperature	MAX.TEMP. > 30^0C, MIN.TEMP. < 10^0C	MAX.TEMP. > 35^0C, MIN.TEMP. < 15^0C	MAX.TEMP. > 30^0C, MIN.TEMP. < 15^0C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	JASSIDS, APHIDS,THRIPS, SEEDLING CUTTING INSECTS	JASSIDS, APHIDS,THRIPS, MITES, SPOTTED BOLL WORM	WHITE FLY, PINK BOLL WORM, RED MITES
	Weather	CLOUDY AND HUMID WEATHER	FROST AND HUMID WEATHER	FROST AND HUMID WEATHER
	Diseases		WILT, ALTERNARIA, LEAF SPOT, ANTHRAXNOSE	ALTERNARIA, LEAF SPOT, ANTHRAXNOSE, BLACK ARM
	Weather		HUMID CLIMATE AND CLOUDY WEATHER	HUMID CLIMATE AND CLOUDY WEATHER

Normal phase wise water requirement(mm)								**124.2**					**148.5**					**291.6**							**195.6**							**104.8**									
Weekly normal weather	*Rainfall(mm) total*					0.5	3.4	7.4	24.8	37.1	30.7	56.4	41.8	45.9	43.0	37.6	36.7	55.4	51.4	34.0	42.9	24.6	17.4	22.6	23.7	15.3	11.5	3.3	2.6	5.2	5.2	1.0	7.4	6.6	0.4				
	Max. temp. ^0C					40.4	40.0	39.3	37.3	36.5	34.9	32.6	31.5	30.9	30.6	30.5	30.0	29.0	29.0	29.4	29.1	30.0	31.2	32.6	32.5	33.0	33.8	34.1	33.2	32.6	31.9	31.6	30.8	29.6	29.4				
	Min. temp. ^0C					26.0	25.9	25.8	25.1	24.8	24.5	23.7	23.5	23.3	23.2	23.0	22.9	22.6	22.4	22.2	22.3	22.0	22.1	22.5	22.4	22.0	21.3	20.8	19.6	18.8	17.9	17.4	16.4	15.7	15.5				
	Sunshine hours					10.1	10.0	10.0	8.9	8.3	6.9	4.9	4.4	4.2	4.2	4.4	3.0	3.2	4.0	4.5	5.0	4.7	6.7	7.4	7.4	8.0	8.9	9.5	9.6	9.8	9.3	9.3	9.2	9.1	9.1				

Life history and mean dates of important epochs of crop growth



Standard weeks			18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	
Months			MAY				JUNE					JULY					AUGUST							SEPTEMBER					OCTOBER					NOVEMBER				DECEMBER		

D.D.G.M.(AGRIMET), I.M.D., PUNE 2002.

CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP :** GROUNDNUT (KHARIF) **VARIETY** : JL-24,AK-12,24,SB-11,TAG-24 **SOIL :** HEAVY,MEDIUM BLACK TO RED LATERITE

DISTRICTS : NAGPUR, AMARAVATI, DHULE **NON-IRRIGATED** **DURATION :** 120 - 140 DAYS

Weather warnings	*Rain*	> 100 MM/DAY	> 100 MM/DAY	> 75 MM/DAY
	Duration of wet spell	30 MM FOR 10 DAYS	30 MM FOR 10 DAYS	25 MM FOR 10 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	18 DAYS	15 DAYS	15 DAYS
	High winds			
	Temperature	MAX. TEMP. > 35^{0}C, MIN.TEMP. < 20^{0}C	MAX. TEMP. > 35^{0}C, MIN.TEMP. < 25^{0}C	MAX. TEMP. > 30^{0}C, MIN.TEMP. < 25^{0}C
	Hail Storm			
Weather conditions favourable for incidence of pests and diseases	*Pests*	JASSIDS, APHIDS, LEAF MINER	JASSIDS, APHIDS, LEAF MINER	JASSIDS, APHIDS, LEAF MINER
	Weather	HIGH HUMIDITY, EXCESS RAINFALL	HIGH HUMIDITY, EXCESS RAINFALL	HIGH HUMIDITY, EXCESS RAINFALL
	Diseases	WHITE & ROOT ROT	TIKKA	TIKKA
	Weather	CLOUDY WEATHER, CONTINUOUS RAINFALL	CLOUDY WEATHER, CONTINUOUS RAINFALL	CLOUDY WEATHER, CONTINUOUS RAINFALL

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP** : GROUNDNUT (KHARIF) **VARIETY** : JL-24, SB-11, KARAD-4,11 **SOIL** : MEDIUM TO HEAVY BLACK
DISTRICTS : KOLHAPUR, SANGLI, AHMEDNAGAR, SATARA **NON-IRRIGATED** **DURATION** : 90 - 120 DAYS

Weather warnings	*Rain*	> 50 MM/DAY	> 100 MM/DAY	> 50 MM/DAY
	Duration of wet spell			
	Cloudy weather			
	Drought	10 DAYS	15 DAYS	15 DAYS
	High winds			
	Temperature	MAX. TEMP > 35°C, MIN.TEMP < 20°C	MAX. TEMP > 35°C, MIN.TEMP < 20°C	MAX. TEMP > 35°C, MIN.TEMP < 20°C
	Hail Storm			
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS, LEAF MINER, WHITE GRUBS; LEAF ROLLER	LEAF MINER, WHITE GRUBS, LEAF ROLLER, APHIDS, LEAF EATING CATERPILLAR	LEAF FOLDER, LEAF ROLLER, WHITE GRUBS, APHIDS, LEAF MINER
	Weather	CLOUDINESS, HIGH RAINFALL	HIGH RAINFALL, DRY SPELL	HIGH RAINFALL, DRY SPELL
	Diseases	WILT, TIKKA, RUST, ROOT ROT	TIKKA, WILT, RUST	TIKKA, WILT, RUST
	Weather	CLOUDINESS, EXCESS RAINFALL	CLOUDINESS, EXCESS RAINFALL	HIGH HUMIDITY,CLOUDINESS

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : KOLHAPUR, PARBHANI, PUNE, SINDHUDURG

CROP : GROUNDNUT (RABI)
NON-IRRIGATED

VARIETY : SB-11
DURATION : 90-135 DAYS

SOIL : MEDIUM TO LIGHT BLACK

Weather warnings	*Rain*			
	Duration of wet spell			
	Cloudy weather			CLOUDY WEATHER
	Drought	7 DAYS	8-10 DAYS	8-10 DAYS
	High winds			
	Temperature	MAX.TEMP. > 31^{0}C, MIN.TEMP. < 20^{0}C	MAX.TEMP. > 34^{0}C, MIN.TEMP. < 20^{0}C	MAX.TEMP. > 38^{0}C, MIN.TEMP. < 20^{0}C
	Hail Storm			
Weather conditions favourable for incidence of s and diseases	*Pests*	APHIDS	LEAF ROLLER,JASSIDS	
	Weather	CLOUDY WEATHER	CLOUDY WEATHER	
	Diseases		TIKKA ,LEAF SPOTS,RUST	TIKKA,BUD NECROSIS
	Weather		CLOUDINESS	CLOUDINESS

| | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP :** GRAM **VARIETY** : CHAFA, CHAFA G-1 **SOIL :** MEDIUM TO DEEP BLACK
DISTRICTS : AHMEDNAGAR, SOLAPUR, KOLHAPUR **IRRIGATED/NON-IRRIGATED** **DURATION :** 110-115 DAYS

Weather warnings	*Rain*			
	Duration of wet spell			
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	40 DAYS	40 DAYS	40 DAYS
	High winds			> 15-20 KM/HR
	Temperature	MIN.TEMP. < 5^{0}C	MIN.TEMP. < 5^{0}C	MIN.TEMP. < 5^{0}C
	Hail Storm			
Weather conditions favourable for incidence of pests and diseases	*Pests*	POD BORER, HELIOTHIS	POD BORER, HELIOTHIS	POD BORER, HELIOTHIS
	Weather	HIGH HUMIDITY, CLOUDINESS	HIGH HUMIDITY, CLOUDINESS	HIGH HUMIDITY, CLOUDINESS
	Diseases	WILT		
	Weather	EXCESS SOIL MOISTURE		

| **Normal phase wise water requirement(mm)** | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : WARDHA, YAVATMAL, NAGPUR, PARBHANI

CROP : GRAM
IRRIGATED/NON-IRRIGATED

VARIETY : CHAFA G-5, ICGS-2,5, BDN 9-3
DURATION : 95 - 120 DAYS

SOIL : HEAVY, MEDIUM OR LIGHT BLACK AND RED LATERITE

Weather warnings	*Rain*	> 50 MM/DAY	> 50 MM/DAY	> 50 MM/DAY
	Duration of wet spell	20 MM FOR 3 OR 4 DAYS	20 MM FOR 3 OR 4 DAYS	20 MM FOR 3 OR 4 DAYS
	Cloudy weather	CLOUDY WEATHER	CLOUDY WEATHER	CLOUDY WEATHER
	Drought	13 DAYS	17 DAYS	9 DAYS
	High winds			
	Temperature	MAX. TEMP. > 30^{0}C, MIN.TEMP. < 20^{0}C	MAX. TEMP. > 30^{0}C, MIN.TEMP. < 20^{0}C	MAX. TEMP. > 30^{0}C, MIN.TEMP. < 20^{0}C
	Hail Storm		HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*		POD BORER, HELIOTHIS	POD BORER, HELIOTHIS
	Weather		CLOUDY AND HUMID WEATHER	CLOUDY AND HUMID WEATHER
	Diseases	WILT,LEAF CURLING VIRUS,FUSARIUM WILT	LEAF CURLING VIRUS,FUSARIUM WILT,SMUT	SMUT
	Weather	CONTINUOUS RAIN, CLOUDY & VERY COLD WEATHER	CLOUDY WEATHER	HIGH HUMIDITY

| **Normal phase wise water requirement(mm)** | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP :** GRAM **VARIETY** : PHULE-G-12, CHAFA G-5, N-59, 31, VIKAS, VISWAS, LOCAL **SOIL :** LIGHT, MEDIUM AND HEAVY BLACK

DISTRICTS : PUNE, DHULE, THANE **IRRIGATED/NON-IRRIGATED** **DURATION :** 95-120 DAYS

Weather warnings	*Rain*	> 100-150 MM/DAY	> 100-150 MM/DAY	> 100-150 MM/DAY
	Duration of wet spell	100 MM FOR 2 DAYS	100 MM FOR 2 DAYS	100 MM FOR 2 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought			
	High winds	> 60 KM/HR	> 60 KM/HR	> 60 KM/HR
	Temperature	MAX. TEMP. > 25^{0}C, MIN.TEMP. < 10^{0}C	MAX. TEMP. > 25^{0}C, MIN.TEMP. < 10^{0}C	MAX. TEMP. > 35^{0}C, MIN.TEMP. < 15^{0}C
	Hail Storm		HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	POD BORER	POD BORER	POD BORER
	Weather	CLOUDY WEATHER	CLOUDY WEATHER	CLOUDY WEATHER
	Diseases	ROOT ROT		
	Weather	HEAVY RAINFALL AND CLOUDINESS		

Normal phase wise water requirement(mm)		52.8	60.9	119.3	135.9	77.4

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : RATNAGIRI, SINDHUDURG, THANE, RAIGAD

CROP : PADDY (KHARIF) **NON-IRRIGATED**

VARIETY : KARJAT-184, RATNA, JAYA, R.P. 4, 14
DURATION : 135 - 150 DAYS

SOIL : LATERITE

Weather warnings	*Rain*	>100 MM / DAY	>125 MM / DAY	>100 MM / DAY
	Duration of wet spell	> 150 MM FOR 3 DAYS	> 125 MM FOR 4 DAYS	> 100 MM FOR 4 DAYS
	Cloudy weather			
	Drought	5 DAYS	8 DAYS	10 DAYS
	High winds	> 35 KM/HR	> 35 KM/HR	> 35 KM/HR
	Temperature	MAX. TEMP. > 32^{0}C MIN. TEMP.< 15^{0}C	MAX. TEMP. > 25^{0}C MIN. TEMP.< 22^{0}C	MAX. TEMP. > 28^{0}C MIN. TEMP.< 24^{0}C
	Hail Storm			HAIL STROM
Weather conditions favourable for incidence of pests and diseases	*Pests*	ARMY WORM, BPH, STEM BORER, LEAF FOLDER	ARMY WORM, BPH, STEM BORER,CASE WORM	LEAF ROLLER, ARMY WORM, BPH, STEM BORER, LEAF FOLDER
	Weather	HIGH TEMP., HIGH HUMIDITY, CLOUDY	CLOUDY, HIGH HUMIDITY	7-8 DAYS CLOUDY WITH BRIGHT SUNSHINE
	Diseases	BLAST, BACTERIAL BLIGHT, BLB	LEAF BLIGHT, BLB	BLAST, BACTERIAL BLIGHT, BLB
	Weather	HUMIDITY, CLOUDY WEATHER	HIGH HUMIDITY , DRY SPELL, CLOUDY WEATHER	CLOUDY WEATHER

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP** : PADDY (KHARIF) **VARIETY** : SKALD- 1&6, SIND -75, SIUDEWALI-75, SWARNA, WHITE LUCHAIL, SUREKHA, JAYA, SOUMASURI **SOIL** : MEDIUM BLACK TO RED & YELLOW LATERITE

DISTRICTS : BHANDARA, CHANDRAPUR, GADCHIROLI, NAGPUR **IRRIGATED/NON-IRRIGATED** **DURATION** : 115-140 DAYS

Weather warnings	*Rain*	> 150 MM/DAY	> 150 MM/DAY	> 100 MM/DAY
	Duration of wet spell	> 40 MM FOR 10 DAYS	> 40 MM FOR 10 DAYS	> 20 MM FOR 10 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	20 DAYS	20 DAYS	30 DAYS
	High winds			
	Temperature	MAX. TEMP. > 41°C, MIN. TEMP. < 20°C	MAX. TEMP. > 35°C, MIN. TEMP. < 20°C	MAX. TEMP. > 40°C, MIN. TEMP. < 20°C
	Hail Storm			
Weather conditions favourable for incidence of pests and diseases	*Pests*	ARMY WORM, LEAF ROLLER, APHIDS, STEM BORER, GALL FLY, CATERPILLAR, GRASS HOPPER	ARMY WORM, LEAF ROLLER, STEM BORER, GALL MIDGE, GALL FLY, BPH	BPH, ARMY WORM, LEAF FOLDER, CATERPILLAR
	Weather	HIGH HUMIDITY	LOW RH, HIGH TEMP.	CLOUDY WEATHER, LOW RH, HIGH TEMP.
	Diseases	ZINC DEFICIENCY, BLAST, ROOT ROT	BLAST, BLB, BROWN SPOT, HELMINTHOSPORIUM	NECK BLAST, FALSE SMUT, BLAST, BLB, HELMINTHOSPORIUM
	Weather	CLOUDY WEATHER, HIGH TEMP., HIGH RH	CLOUDY WEATHER, HIGH TEMP., HIGH HUMIDITY	CLOUDY WEATHER WITH CONTINUOUS RAIN

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP** : SAFFLOWER **VARIETY** : BHIMA, TARA **SOIL** : MEDIUM TO DEEP BLACK
DISTRICTS : AHMEDNAGAR, PUNE, SOLAPUR **NON-IRRIGATED** **DURATION** : 130-135 DAYS

Weather warnings	*Rain*	>100- 150 MM/DAY	>100- 150 MM/DAY	>100- 150 MM/DAY
	Duration of wet spell	> 40 MM FOR 3DAYS	> 60 MM FOR 5 DAYS	> 50 MM FOR 10 DAYS
	Cloudy weathe.		CLOUDY WEATHER	
	Drought			
	High winds			>15-20 KM/HR
	Temperature	MAX. TEMP. > 40°C, MIN. TEMP. < 15°C	MAX. TEMP. > 40°C, MIN. TEMP. < 15°C	MAX. TEMP. > 40°C, MIN. TEMP. < 15°C
	Hail Storm		HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	LEAF SPOT, APHIDS	LEAF SPOT, APHIDS, CAPSULES BORER	LEAF SPOT, APHIDS, CAPSULES BORER
	Weather	CLOUDINESS, LOW TEMP., HUMID CLIMATE	CLOUDINESS, LOW TEMP., HUMID CLIMATE	CLOUDINESS, LOW TEMP., HUMID CLIMATE
	Diseases	LEAF SPOT, APHIDS	LEAF SPOT, WILT	LEAF SPOT, ALTERNARIA, LEAF SPOT
	Weather	CLOUDINESS	CLOUDINESS	CLOUDINESS

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : BULDHANA, PARBHANI, AURANGABAD

CROP : SAFFLOWER
NON-IRRIGATED

VARIETY : TARA, BHIMA, S-4
DURATION : 125-140 DAYS

SOIL : LIGHT TO HEAVY BLACK

Weather warnings	*Rain*	> 50 MM/DAY	> 70 MM/DAY	> 70-80 MM/DAY
	Duration of wet spell	> 40 MM FOR 3 DAYS	> 50 MM FOR 3 DAYS	> 50 MM FOR 3 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	15 DAYS	20 DAYS	20 DAYS
	High winds	> 40 KM / HR	> 40 KM / HR	> 40 KM / HR
	Temperature	MAX. TEMP. > 30°C, MIN. TEMP. < 20°C	MAX. TEMP >35°C, MIN. TEMP. < 17°C	MAX. TEMP. > 35°C, MIN. TEMP. < 20°C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS	APHIDS	APHIDS
	Weather	HIGH HUMIDITY, LOW TEMP	HIGH HUMIDITY, LOW TEMP	HIGH HUMIDITY, LOW TEMP
	Diseases	FUSARIUM WILT, RR, ALB	FUSARIUM WILT, RR, ALB	FUSARIUM WILT, RR, ALB
	Weather	CLOUDINESS	CLOUDINESS	CLOUDINESS

| **Normal phase wise water requirement(mm)** | | | **75.4** | | | | | | | **131.1** | | | | | | | **115.9** | | | | **53.5** | | | | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP :** BAJRA (KHARIF) **VARIETY** : BK-560,MH-179,182,168,143, ICTP-8203, ICMS-7703,WCC-75, ICCMN-87901 **SOIL :** MEDIUM BLACK

DISTRICTS : DHULE, NASIK, AHMEDNAGAR, AURANGABAD, JALNA **NON-IRRIGATED** **DURATION :** 85-110 DAYS

Weather warnings	*Rain*	> 60 MM/DAY	> 80 MM/DAY	> 100 MM/DAY
	Duration of wet spell	> 40 MM FOR 3 DAYS	> 50 MM FOR 3 DAYS	> 50 MM FOR 3 DAYS
	Cloudy weather	CLOUDY WEATHER	CLOUDY WEATHER	CLOUDY WEATHER
	Drought	15 DAYS	20 DAYS	20 DAYS
	High winds	> 45 KM/HR	> 45 KM/HR	> 45 KM/HR
	Temperature	MAX TEMP. > 30^0C MIN TEMP. < 10^0C	MAX TEMP.> 35^0C MIN TEMP. < 15^0C	MAX TEMP. > 35^0C MIN TEMP .< 15^0C
	Hail Storm	HAIL STROM	HAIL STROM	HAIL STROM
Weather conditions favourable for incidence of pests and diseases	*Pests*	BLISTER, BEETLE	SHOOT FLY, STEM BORER, EAR HEAD BUG	SHOOT FLY, STEM BORER, EAR HEAD BUG
	Weather	HUMIDITY, CLOUDINESS	HUMIDITY, CLOUDINESS	HUMIDITY, CLOUDINESS
	Diseases	DOWNY MILDEW, ERGOT	DOWNY MILDEW, ERGOT	DOWNY MILDEW, ERGOT
	Weather	HIGH HUMIDITY, CLOUDINESS	HIGH HUMIDITY, CLOUDINESS	HIGH HUMIDITY, CLOUDINESS

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : PUNE, SATARA, SANGLI, SOLAPUR

CROP : BAJRA (KHARIF) **VARIETY** : WBH-4, ICTP-8203, WCC-75, WCC-25, MBH-104
NON-IRRIGATED **DURATION** : 85-110 DAYS

SOIL : LIGHT TO HEAVY BLACK

Weather warnings	*Rain*	> 75 MM/DAY	> 75-100 MM/DAY	> 100 MM/DAY
	Duration of wet spell	> 5 MM FOR 12 DAYS	> 10 MM FOR 10 DAYS	> 25 MM FOR 7 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought		30 DAYS	
	High winds			
	Temperature	MAX TEMP. > 40°C MIN TEMP. < 15°C	MAX TEMP.> 35°C MIN TEMP. < 15°C	MAX TEMP. > 35°C MIN TEMP .< 15°C
	Hail Storm			HAIL STROM
Weather conditions favourable for incidence of pests and diseases	*Pests*	BLISTER, BEETLE	BLISTER, BEETLE	BLISTER, BEETLE
	Weather	HIGH HUMIDITY, CLOUDY	HIGH HUMIDITY, CLOUDY	HIGH HUMIDITY, CLOUDY
	Diseases	DOWNY MILDEW, ERGOT	DOWNY MILDEW, ERGOT	DOWNY MILDEW, ERGOT
	Weather	CLOUDY, HUMIDITY	CLOUDY, RAINFALL	CLOUDY, HUMIDITY

Normal phase wise water requirement(mm)		**73.0**	**69.8**	**149.5**	**32.1**

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : AHMEDNAGAR,OSMANABAD, SOLAPUR,PARBHANI

CROP : TUR / ARHAR (KHARIF)
NON-IRRIGATED

VARIETY : BDN-1, 2
DURATION : 155-180 DAYS

SOIL : LIGHT TO MEDIUM BLACK

Weather warnings	*Rain*				
	Duration of wet spell				60-70 MM FOR 5 DAYS
	Cloudy weather			CLOUDY WEATHER	
	Drought	45 DAYS		45 DAYS	45 DAYS
	High winds				>15-20 KM/HR
	Temperature	MAX.TEMP. > 35°C, MIN.TEMP. < 10°C		MAX.TEMP. > 35°C, MIN.TEMP. < 10°C	MAX.TEMP. > 35°C, MIN.TEMP. < 10°C
	Hail Storm				
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS, LEAF ROLLER, LEAF EATING CATERPILLAR		CATERPILLAR, HAIRY CATERPILLAR, HELIOTHIS, POD FLY	TUR POD BORER
	Weather	LOW HUMIDITY		CLOUDINESS AND HUMID CLIMATE	CLOUDINESS AND HUMID CLIMATE
	Diseases	WILT		WILT, MOSAIC	WILT, MOSAIC
	Weather	EXCESS RAINFALL, CLOUDINESS & WATER LOGGING CONDITION		EXCESS RAINFALL, CLOUDINESS & WATER LOGGING CONDITION	EXCESS RAINFALL, CLOUDINESS & WATER LOGGING CONDITION

| **Normal phase wise water requirement(mm)** | | | | | | | **105.5** | | | | | | **134.4** | | | | | **235.1** | | | | | | **186.1** | | | | | **116.2** | | | | | | |

CROP WEATHER CALENDAR

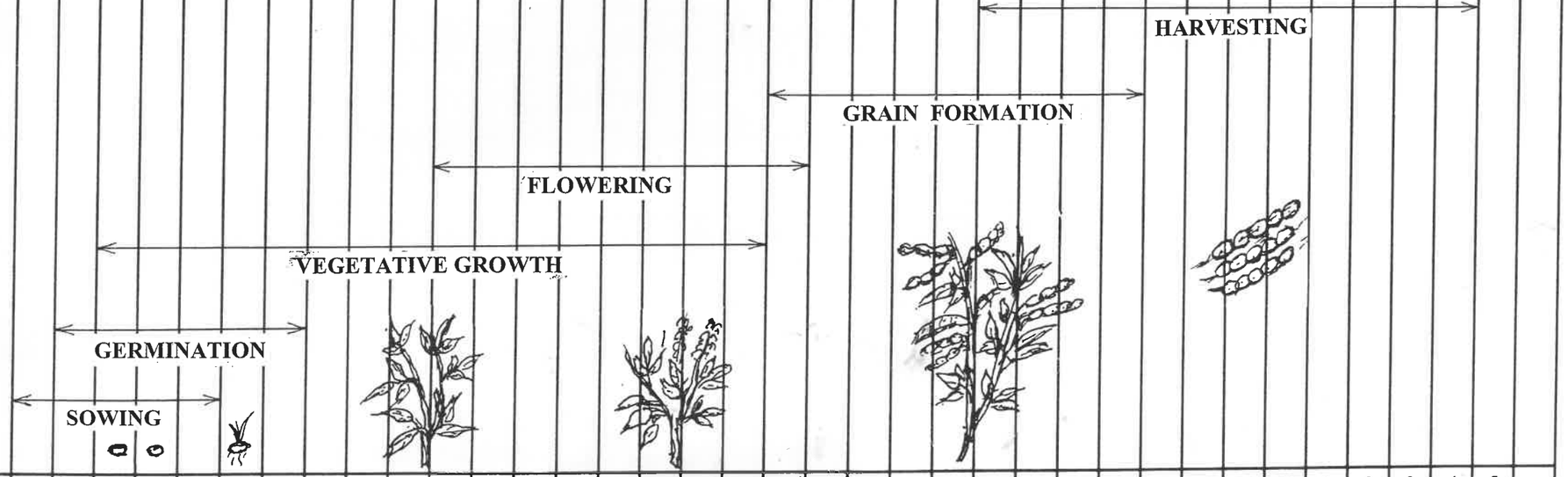
STATE : MAHARASHTRA **CROP :** TUR / ARHAR (KHARIF) **VARIETY** : BDN-1,2,T-VISHAKA,TAT-10, ICPL-87,HYDRABAD-135, C-11 **SOIL :** MEDIUM TO HEAVY BLACK AND LIGHT RED LATERITE

DISTRICTS : BULDHANA, AMARAVATI, AURANGABAD, NAGPUR **NON-IRRIGATED** **DURATION :** 155-180 DAYS

Weather warnings	*Rain*	> 75 MM / DAY	> 100 MM / DAY	> 100 MM / DAY
	Duration of wet spell	> 40 MM FOR 3 DAYS	> 50 MM FOR 3 DAYS	> 50 MM FOR 3 DAYS
	Cloudy weather		CLOUDY WEATHER	CLOUDY WEATHER
	Drought	> 15 DAYS	> 20 DAYS	> 15 DAYS
	High winds	> 15 KM/HR	> 15 KM/HR	> 30 KM/HR
	Temperature	MAX.TEMP. > 30°C, MIN.TEMP. < 10°C	MAX.TEMP. > 40°C, MIN.TEMP. < 15°C	MAX.TEMP. > 40°C, MIN.TEMP. < 15°C
	Hail Storm		HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	APHIDS, JASSIDS	LEAF ROLLER, LEAF MINER, BOLL WORM	POD FLY, POD BORER, HELIOTHIS, BOLL WORM
	Weather	HIGH HUMIDITY, CLOUDINESS	HIGH HUMIDITY, CLOUDINESS	HIGH HUMIDITY, CLOUDINESS
	Diseases	ROOT ROT	WILT, MOSAIC, ANGULAR LEAF SPOT	WILT, STEM BLAST, ANGULAR LEAF SPOT
	Weather	CONTINOUS WET CONDITION		DRY SPELL

Normal phase wise water requirement(mm)							**56.8**		**131.8**					**298.8**								**213.9**						**82.4**																	
Weekly normal weather	*Rainfall(mm) total*		8.9	21.4	38.7	42.8	56.7	49.7	50.6	43.5	42.5	53.0	62.5	52.9	49.3	52.2	40.2	21.6	29.4	25.9	23.7	15.8	3.6	7.4	5.2	3.8	1.3	8.2	4.7	1.4	1.7	0.9	3.5	1.9	1.9	1.4	2.8							
	Max. temp. ^0C		40.1	38.1	36.7	34.5	32.2	31.3	30.4	30.4	30.5	30.0	28.8	29.1	29.5	29.2	29.9	30.9	32.1	32.2	32.1	32.6	32.6	31.8	31.2	30.5	30.3	29.7	28.8	28.3	28.0	28.2	27.8	27.6	28.0	28.3	28.8							
	Min. temp. ^0C		27.1	25.9	25.0	24.5	23.7	23.3	23.1	23.1	23.1	22.8	22.4	22.5	22.3	22.3	22.1	22.3	22.7	22.5	22.2	21.5	21.0	19.7	18.9	18.2	17.7	16.8	16.2	15.4	14.7	14.5	14.5	14.4	14.9	15.0	15.4							
	Sunshine hours		9.1	8.3	7.3	6.1	4.5	3.4	3.6	3.7	3.7	3.0	2.9	3.5	3.8	4.6	4.0	5.6	6.3	7.1	7.8	8.5	9.2	9.0	9.4	9.2	9.6	9.2	9.2	9.1	8.9	9.0	9.4	8.8	9.3	9.4	9.6							

Life history and mean dates of important epochs of crop growth



Standard weeks	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	1	2	3	4	5
Months	JUNE					JULY				AUGUST					SEPTEMBER				OCTOBER				NOVEMBER				DECEMBER				JANUARY				

D.D.G.M.(AGRIMET), I.M.D., PUNE 2002.

CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP** : SUNFLOWER (KHARIF) **VARIETY** : EC - 68414 **SOIL** : LIGHT TO MEDIUM BLACK

DISTRICTS : AHMEDNAGAR, SOLAPUR **IRRIGATED/NON-IRRIGATED** **DURATION** : 100-110 DAYS

Weather warnings	*Rain*			
	Duration of wet spell		> 60-70 MM FOR 4-5 DAYS	> 60-70 MM FOR 5-7 DAYS
	Cloudy weather			
	Drought	15 DAYS	15 DAYS	15 DAYS
	High winds			
	Temperature	MAX. TEMP. > 30^{0}C, MIN. TEMP. < 8^{0}C	MAX. TEMP. > 30^{0}C, MIN. TEMP. < 8^{0}C	MAX. TEMP. > 30^{0}C, MIN. TEMP. < 8^{0}C
	Hail Storm			
Weather conditions favourable for incidence of pests and diseases	*Pests*	JASSIDS, APHIDS, LEC, HAIRY CATERPILLAR	JASSIDS, APHIDS, LEC, HAIRY CATERPILLAR	JASSIDS, APHIDS, LEC, HAIRY CATERPILLAR
	Weather	CLOUDY AND HUMID WEATHER	CLOUDY AND HUMID WEATHER	CLOUDY AND HUMID WEATHER
	Diseases	RHYZOCTONIA, WILT, RUST, LB, ROOT ROT	RHYZOCTONIA, WILT, RUST, LB, ROOT ROT	RHYZOCTONIA, WILT, RUST, LB, ROOT ROT
	Weather	CLOUDY & HUMID CLIMATE WITH WATER LOGGING	CLOUDY & HUMID CLIMATE WITH WATER LOGGING	CLOUDY & HUMID CLIMATE WITH WATER LOGGING
Normal phase wise water requirement(mm)		**65.8**	**107.1**	**158.3** **41.8**

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CROP WEATHER CALENDAR

STATE : MAHARASHTRA **CROP :** SOYABEAN (KHARIF) **VARIETY** : PK-472,MAC-5,13,58, MONETA **SOIL :** MEDIUM HEAVY BLACK AND RED LATERITE

DISTRICTS : AMARAVATI, KOLHAPUR, NAGPUR **NON-IRRIGATED** **DURATION :** 85-125 DAYS

Weather warnings	*Rain*	>75-100 MM / DAY	> 100-150 MM / DAY	> 90-150 MM / DAY
	Duration of wet spell	50 MM FOR 4 DAYS	75 MM FOR 12 DAYS	2-3 MM FOR 6 DAYS
	Cloudy weather		CLOUDY WEATHER	
	Drought	8-15 DAYS	8-20 DAYS	8-20 DAYS
	High winds			
	Temperature	MAX.TEMP. > 35^{0}C, MIN.TEMP. < 20^{0}C	MAX.TEMP. > 38^{0}C, MIN.TEMP. < 20^{0}C	MAX.TEMP. > 38^{0}C, MIN.TEMP. < 20^{0}C
	Hail Storm			HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	STEM FLY, LEC, APHIDS, JASSIDS	LEC, LEAF FOLDERS, LEAF MINER, GIRDLE BEETLE	GLR, POD BORER, STEM FLY, LEAF FOLDER
	Weather	HIGH TEMP., DRY SPELL, LOW HUMIDITY	HIGH TEMP., DRY SPELL, LOW HUMIDITY	HIGH TEMP., DRY SPELL, LOW HUMIDITY
	Diseases	WILT	TIKKA	SOYABEAN MOSAIC
	Weather	HIGH TEMPERATURE	HIGH HUMIDITY AND CLOUDINESS	HIGH HUMIDITY AND CLOUDINESS

| **Normal phase wise water requirement(mm)** | | | | | | | | | | **46.9** | | | | **92.1** | | | | **137.7** | | | | **76.4** | | | | **45.7** | | | | | | | | | |

CROP WEATHER CALENDAR

STATE : MAHARASHTRA
DISTRICTS : PUNE, SATARA

CROP : SUGARCANE
IRRIGATED

VARIETY : CO-7219, 7527,740,419,7125,720, COM-7125
DURATION : 480-490 DAYS

SOIL : MEDIUM TO HEAVY BLACK

Weather warnings	*Rain*	> 50 MM/DAY	> 100 MM/DAY	> 80 MM/DAY
	Duration of wet spell	> 30 MM FOR 5 DAYS	> 50 MM FOR 5 DAYS	> 20 MM FOR 5 DAYS
	Cloudy weather			CLOUDY WEATHER
	Drought	15-20 DAYS	20-30 DAYS	25-30 DAYS
	High winds	> 40-50 KM/HR	> 30-40 KM/HR	> 30-40 KM/HR
	Temperature	MAX.TEMP. > 35^{0}C, MIN.TEMP. < 12^{0}C	MAX.TEMP. > 35^{0}C, MIN.TEMP < 16^{0}C	MAX.TEMP. > 35^{0}C, MIN.TEMP. < 20^{0}C
	Hail Storm	HAIL STORM	HAIL STORM	HAIL STORM
Weather conditions favourable for incidence of pests and diseases	*Pests*	SHOOT BORER	SCALE INSECTS	STEM BORER
	Weather	DRY SPELL, HOT WEATHER	DRY SPELL, HOT WEATHER	DRY SPELL, HOT WEATHER
	Diseases	BLACK SMUT, SHOOT BORER	BLACK SMUT	BLACK SMUT
	Weather	HOT WEATHER, CLOUDINESS	HOT WEATHER, CLOUDINESS	HOT WEATHER, CLOUDINESS
Normal phase wise water requirement(mm)		**		

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